

System Report: Image-to-Video Semantic Retrieval via Object Detection

Course: Artificial Intelligence — Spring 2026

Assignment: 2

1. Detector Choice and Configuration

For the object detection backbone, the system utilizes the YOLOv8 Nano Segmentation model (yolov8n-seg.pt). This model balances computational efficiency with high accuracy, making it well-suited for processing video frames.

The base model was fine-tuned on the Ultralytics Car Parts Segmentation Dataset. The training was conducted for 30 epochs using an image size of 640 pixels. While the model natively predicts segmentation masks, the retrieval system operates on bounding boxes to simplify indexing. During inference, the segmentation masks are bypassed, and the spatial coordinates are extracted using the model's xyxy bounding box outputs.

To filter out false positives and ensure high-quality semantic matching, a strict confidence threshold was implemented. Only detections satisfying the condition $\text{conf} > 0.50$ are recorded in the system.

2. Video Sampling Strategy and Data Engineering

The video corpus processing pipeline is designed for speed and minimal storage overhead.

Video Acquisition and Frame Extraction

The source video is downloaded using yt-dlp. To optimize the download, only the video stream is fetched, completely bypassing the audio stream which is irrelevant for visual retrieval. The video is then passed to ffmpeg for frame extraction. Frames are sampled at a rate of 1 frame every 5 seconds ($\text{fps} = 1/5$) using a direct stream copy approach to avoid the severe computational penalty of re-encoding.

Temporal Mapping and Indexing

Because the extraction pipeline naturally outputs frames sequentially starting at frame_0001.jpg, the system natively aligns with a 1-based index approach. The chronological timestamp (in seconds) for each extracted frame is calculated using the formula:

$$t_{\text{sec}} = (n - 1) \times 5$$

where n is the frame index.

Once the frames are passed through the fine-tuned YOLOv8 model, the results are compiled into a highly structured Pandas DataFrame and exported as video_index.parquet. The schema serves as the sole interface between the detection and retrieval logic, containing:

- video_id
- frame_index
- timestamp_sec
- class_label
- bounding_box
- confidence_score

3. Image-to-Video Matching Logic

The retrieval system uses a shared representation space, meaning the exact same detector used to index the video is used to analyze the query images from the Hugging Face aegean-ai/rav4-exterior-images dataset.

Query Analysis

When a query image is submitted, the model identifies all car components present. The unique class labels for all detections exceeding the 0.50 confidence threshold are collected into a query set. If no parts are confidently detected, the system gracefully terminates the search.

Semantic Matching and Temporal Grouping

The system loads the Parquet index and filters it exclusively for rows where the class_label matches the items in the query set. To construct contiguous video clips from individual frames, the system groups the matching records by class and sorts them chronologically.

It then calculates the temporal gap (Δt) between consecutive frames. Because the base sampling rate is 5 seconds, any gap where $\Delta t \leq 6$ seconds is treated as a continuous visual block. If the gap exceeds 6 seconds, the system logically splits the sequence into a new, distinct video segment.

Output Formatting

For each contiguous block, the system aggregates the data to produce the final required deliverables:

- start_timestamp: The earliest t_sec in the block.
- end_timestamp: The latest t_sec in the block.
- class_label: The specific component being queried.
- number_of_supporting_detections: The total count of matching frames within that block.

The final clips are returned sorted by their robustness, prioritizing segments with the highest number of supporting detections.

4. Failure Cases and Limitations

While the current pipeline effectively retrieves semantic matches, there are a few inherent limitations:

- **Architectural Overhead:** The system utilizes a fine-tuned segmentation model but only leverages the bounding box outputs for indexing. This means the model wastes computational resources calculating pixel-perfect masks during the video processing phase, slowing down the overall indexing speed compared to using a pure bounding box model (yolov8n.pt).
- **Temporal Fragmentation:** The strict $\text{conf} > 0.50$ threshold can lead to flickering. If an object is momentarily occluded, suffers from motion blur, or dips to a 0.49 confidence score for a single frame, the system will instantly break the contiguous block and register two highly fragmented, separate clips instead of one smooth segment.
- **Contextual Blindness:** The matching logic relies entirely on class label overlaps (e.g., retrieving 'wheel'). If a query image focuses on the front-left wheel, the system will currently retrieve clips of any wheel in the video, as it lacks the spatial tracking or relational context needed to distinguish between multiple instances of the same class.